

INF2003: Database Systems

Lab 04: SQL in Python

Objectives

Write SQL statements in Python environment to perform data definition and manipulation operations.

Deliverables

You are required to finish the lab questions listed in the notebook and submit the file with your code and output to Xsite. (Please make a copy in your drive before you work on it)

Background Problem Statement

We learned how to write an object-oriented database in Python in our first lab, without using SQL. We also learned how to code in SQL for database management and operation in the following labs. Given the fact that Python is becoming more and more prevalent, we are more likely to deal with our databases in Python instead of the traditional SQL editors in the future. In this lab, we aim to mix SQL and Python and learn how to code SQL directly in Python, with certain Python library and functions.

Exercise: SQL in Python.

(Note: each question corresponds to a Jupyter cell.)

- Q1. With Python library `sqlite3`, create (with `connect`) a database called “music.sqlite” and delete the table called “Tracks” if it exists. Then, you create a new table called “Tracks” with 2 columns, “title” as TEXT and “plays” as Integer. (You may think deleting Tracks in the beginning is unnecessary. Is this true?)
- Q2. Insert 2 tuples shown below into the table “Tracks”.
(‘We Don’t Talk About Bruno’, 20)
(‘Levitating’, 15)
- Q3. Retrieve all tuples from the table, return them to a cursor, print the existing tuples row by row with a for-loop.
- Q4. Retrieve all tuples from the table, return them to a cursor, print the 1st tuple with `fetchone()`.
- Q5. Retrieve all tuples from the table, return them to a cursor, print all tuples with `fetchall()`. (Do we need a for-loop?)
- Q6. Update the plays of “We Don’t Talk About Bruno” from 20 to 5. Print the latest table.
- Q7. Delete the tuples with plays equal to 5. Print the latest table.
- Q8. 1) Create another database “order.db”.
2) Delete two tables “customers” and “orders” if any.
3) Then, create table “customers” with 2 columns, “customer_id” as Integer as primary key and “customer_name” as TEXT.

4) Create another table “orders” with 3 columns, “order_id” as TEXT as primary key, “customer_id” as Integer, and “amount” as DOUBLE. Meanwhile, specify a foreign key as “customer_id” referring to the “customer_id” in table “customers”.

Q9. Insert the below tuples into table “customers”. Print the latest table.

(1, 'Arya')
(2, 'Jon')
(3, 'Robb')
(4, <Yourself>)

Q10. Insert the below tuples into table “orders”. Print the latest table.

('O1', 1, 19.99)
('O2', 1, 66.66)
('O3', 3, 17.56)
('O4', 4, 88.88)

Q11. Join table “customers” and table “orders” together based on column “customer_id”. Print the table after join with both tables’ data. (Do we have any information about Jon in the output?)

Q12. In the above question, you practiced how to return multiple columns from different tables. Here, return the “customer_name” and “order_id” of all the orders each with an amount greater than 15.

Submission

Format: **ONE (1) notebook file (.ipynb)**

Filename: <lab id>_<student id>_<name>_Lab4.ipynb.

Venue: Dropbox in xSiTe.

Deadline: the following **Monday 11:30pm** when the lab is conducted.

Penalty: A penalty of **10% per day** for late submission will be imposed. A penalty of 100% for this lab will be imposed for the 1st time plagiarism and a penalty of 100% for ALL the labs for the 2nd time plagiarism.